

F1 Deep Dives 2 — seven worlds, the networks, and two new chapters

Sep 28, 2026 · @Mohammad AlFalasi

Scope and method

This round runs deep dives on the seven remaining worlds and on the two chapters that Q1 proposes. Together with F1.5 and F1.6 in [Deep Dives 1](#), that covers every world chapter and the networks unit. Each follows deep dive template v2 and names its groupings by kind, as set out in [F1 Atlas Logic](#).

Each deep dive gives, in the same order:

- the unit's device, its reach (the sub-regions and periods it covers), and the groupings it uses;
- the claims, each with a source and a confidence;
- the object set, checked live for licence, date and provenance;
- the living practice;
- protocol notes;
- the walk;
- Apply by lane;
- success criteria and links out;
- what the deep dive changed.

Assumed. Q1-Q12 and P1-P11 as recommended, so the two new chapters are written in full. If Q1 is declined, their material moves into F1.5, F1.10 and F1.13 as world-hub content.

How the checks were done. Object records were read live from the Cleveland Museum of Art and Metropolitan Museum open APIs on 28 September 2026. Cleveland records carry provenance; Met records do not, so every Met object is marked "provenance unchecked" until its web page is read. Claims come from UNESCO listings, museum essays, scholarly articles and, where noted, encyclopedia summaries that must be replaced before G3. The provenance test is the one from Deep Dives 1: an excavation record, a documented history outside the country of discovery before 17 November 1970, or a documented legal export.

Kept short on purpose. Each deep dive holds only what the unit needs: 10–16 claims and 6–10 objects for the taught path. Everything else found along the way is logged for the world hub rather than squeezed into the 50-minute unit.

F1.13 Networks: objects, materials, skills and people

The unit follows one blue-and-white jar and the cobalt that coloured it, then widens to every route the jar's world touched. The deep dive changed three things. New research points the cobalt route to a different mine, so the mockup's dashed line moves. A famous shipwreck became a case about the ethics of salvage rather than an exemplar. And the Atlantic step now counts people from the scholarly database, as people.

Reach. The Silk Roads, the Indian Ocean, the trans-Saharan routes, the Manila galleons and the Atlantic, from the 2nd century BCE to the 19th century. The Mediterranean belongs to F1.11 and the Pacific voyages to F1.12, each handed on by name. **Groupings used:** networks (the Silk Roads corridor, the Indian Ocean, the Manila galleon route, the Atlantic); an interaction sphere (the Swahili coast towns); polities (Mali, the Yuan); materials (cobalt, gold, silver).

Claims

Claim	Confidence	Source
The Chang'an–Tianshan corridor of the Silk Roads is a 5,000 km network of 33 sites; it took shape between the 2nd century BCE and the 1st century CE and was used until the 16th century for trade, beliefs, science and technology	Documented	UNESCO World Heritage listing
Early Chinese blue-and-white depended on cobalt imported from the West	Documented	PNAS 2020

Claim	Confidence	Source
That cobalt came most likely from Anarak on the Iranian plateau, not from Qamsar near Kashan as previously thought; its signature matches the blue of early Islamic glazed ceramics	Probable	Same study
An Arab-type ship sank off Belitung about 830 with some 60,000 Chinese ceramics; one bowl is dated 826	Documented	Summary citing the Met's John Guy (to replace with the excavation report)
The Smithsonian's Sackler Gallery cancelled its exhibition of the Belitung cargo in 2011 after objections to its commercial salvage	Documented	The Washington Post, 15 December 2011
From the 13th to the 16th century Kilwa's merchants traded gold, silver, pearls, perfumes, Persian earthenware and Chinese porcelain; Kilwa minted its own coins; its Great Mosque's vaults held set-in Chinese porcelain	Documented	UNESCO World Heritage listing
The 1375 Catalan Atlas shows Mansa Musa of Mali holding a gold coin, "the richest and most noble lord" of the region because of its gold	Documented	The atlas sheet, BnF Espagnol 30
The Manila galleons (1565–1815) carried porcelain, silk and ivory from Manila to Acapulco for American silver; Mexican ceramics show the trade's impact most vividly	Documented	The Met, Heilbrunn essay

Claim	Confidence	Source
As much as a third of the silver mined in New Spain and Peru went to Asia	Probable (an estimate)	Same essay
About 12.5 million people were embarked in the trans-Atlantic slave trade between 1501 and 1866	Documented, as an estimate	SlaveVoyages
The first paper mill in Baghdad was set up in 794–95; the story that Chinese prisoners taken at Talas in 751 brought papermaking west is contested	Documented (mill); contested (Talas)	Bloom, <i>Paper Before Print</i> (2001), review
Lapita pottery appears in the Bismarck Archipelago about 3,470–3,250 years ago and reaches Fiji by about 3,130–3,010 years ago	Documented	Archaeology in Oceania, 2012 (handed to F1.12)
Gulf sailors navigated wooden lenj boats by sun, moon and stars and named each wind; the knowledge is passing from use	Documented	UNESCO Urgent Safeguarding List, 2011

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-020	Jar with lion-head handles, Jingdezhen	1300s, Yuan	Cleveland 1962.154, CC0	A New York dealer, sold to Cleveland in 1962	The object carried along the route
WS-021	Footed bowl, probably Kashan	Late 1100s–early 1200s	Cleveland 1915.581, CC0	Bought in 1915	Cobalt blue in Iran before Chinese blue-and-white

ID	Object	Date	Holder and licence	Provenance	Role
WS-022	Basin, Puebla, Mexico, by "Master Potter A"	About 1650	Met 12.3.1, public domain	Given in 1912	The Manila galleon reaches Mexican kilns
WS-006	Block-printed cotton, found in Egypt	1200s–1300s	Cleveland 1929.845, CC0	Acquired 1929	Indian Ocean cotton (shared with F1.5)
WS-010	Benin plaque	1500s–1600s	Cleveland 1999.1, CC0	Deep Dives 1	Rhineland brass on the Atlantic (shared with F1.6)
AS-003	Catalan Atlas, sheet 6	1375	BnF Espagnol 30; public-domain work, reproduction terms to check	—	Gold across the Sahara, as Europe drew it
WS-X05	The Belitung cargo	About 830	Asian Civilisations Museum, Singapore	Commercially salvaged	Excluded as an exemplar; taught in F1.24 as a case about salvage

Living practice. Lenj building and sailing in the Persian Gulf, on UNESCO's urgent list. It is paired with the *Jewel of Muscat*, a sailing reconstruction of the Belitung ship, as a case of reconstruction (F1.28a). It is also the Gulf's first hook into the regional vertical.

Protocol. The Atlantic step follows the sprint 5 rule: people are never drawn as goods. They are counted as people, with a content note, and the step names the brass manillas of F1.6 as what was traded for them. Primary voices come from narratives written by formerly enslaved people, where those texts are in the public domain.

The walk

1. **Guess.** Where did the blue on this jar come from?
2. **The jar.** Jingdezhen, 1300s (WS-020).
3. **The blue.** The Iranian plateau: Anarak probably, Kashan as the older answer. The map redraws the dashed materials line, and the claim shows both.
4. **Blue before China.** The Kashan bowl: Iranian potters used the same cobalt first.
5. **By sea.** The Belitung ship and the argument over its salvage; Kilwa's mosque set with Chinese bowls.
6. **By land.** The Silk Roads corridor; paper's route, and the legend attached to it.
7. **Gold.** Mansa Musa on the Catalan Atlas: West African gold as a European mapmaker drew it.

8. **Across the Pacific.** Silver out, porcelain in, and the Puebla basin.

9. **The Atlantic.** 12.5 million people, the brass traded for them, and their own words.

Apply

Lane	Task
Both	Trace one object's route end to end, in four layers: goods, materials, skills and people
Maker	Name one material in your own work and the route it travels today, with a dated source
Briefer	Write the supply-chain line a brief should require: where the materials come from and who made the thing

How we will know it works. At least 80% of learners separate the four kinds of flow in their trace. At least 70% name a material route with a dated source. In the Atlantic step, learners' own writing uses "enslaved people", never "slaves" as cargo (the writing check in Deep Dives 1 §6).

Links out. F1.5 (paper and printing), F1.6 (brass and gold), F1.9 (cotton), F1.10 (cobalt, paper, Kilwa), F1.12 (Lapita), F1.14 (Puebla copying Chinese style), F1.22 (silver as extraction), F1.24 (the Belitung salvage), F1.28a (the *Jewel of Muscat*), and the two new chapters (the Gulf; the Sogdians and the Mongols).

What changed. The cobalt route is redrawn and its claim updated. The salvage case is added. People in the Atlantic step are counted with the database's estimates. Gulf seafaring becomes the unit's living practice.

F1.12 Australia and the Pacific, read through navigation

This is the chapter with the deepest time and the thinnest open record. People reached Australia at least 50,000 years ago and were grinding axe edges before anyone else in the world; Pacific navigators settled the largest ocean on Earth without writing or metal. Yet the two big open museum APIs offer almost nothing here that is both licensed and appropriate, so the chapter is built on heritage listings, community-led sources and living practice, and it says so.

Reach. Australia (added by Q1), Melanesia, Micronesia and Polynesia, from at least 50,000 years ago to today. **Groupings used:** an archaeological culture (Lapita); communities (Gunditjmara; Marshallese; Native Hawaiian); a network (the Lapita and later voyaging spheres); maker communities (navigators, *ri-metos*); the environment lens (ocean, atoll, lava flow).

Claims

Claim	Confidence	Source
People occupied Madjedbebe, northern Australia, by about 65,000 years ago, with ground ochre, grinding stones and ground-edge hatchet heads	Contested: the excavators' dating, questioned by others who hold to about 50,000 years	Clarkson et al. 2017, <i>Nature</i>; the 2018 reply to critics
A ground-edge axe fragment from the Kimberley is 44,000–49,000 years old, the earliest known anywhere	Documented	Hiscock et al. 2016, <i>Australian Archaeology</i>
The Gunditjmara built and managed stone channels, weirs and dams to trap and farm eels at Budj Bim for at least 6,600 years; the system still works and its knowledge is held today	Documented	UNESCO World Heritage listing
Lapita pottery appears in the Bismarck Archipelago about 3,470–3,250 years ago and reaches Vanuatu and Fiji within a few centuries	Documented	Anderson 2012, <i>Archaeology in Oceania</i>
Marshallese “stick charts” model swells, currents and islands; they were training and planning tools, memorised and left ashore, and were read mainly by their makers	Documented	Smithsonian Ocean

Claim	Confidence	Source
In 1976 the canoe Hōkūle'a sailed from Hawai'i to Tahiti navigated without instruments by Mau Piailug of Satawal; in 1980 Nainoa Thompson repeated it, the first Hawaiian navigator to do so in 600 years	Documented	Polynesian Voyaging Society
Australia has no law against the misuse of traditional symbols, songs and stories; Creative Australia's protocols (Terri Janke, 2002, revised 2007) require consultation, consent and attribution as a condition of funding	Documented	Creative Australia

Objects, and a finding about the pipeline. The Met's search endpoint returned Pacific objects under a public-domain filter that its own records mark "isPublicDomain: false" (a Marshallese *rebbelib* and two Fijian barkcloths). So the Met search filter cannot be trusted for this world; every hit has to be checked at record level, which the template now requires. Cleveland's CC0 set has no barkcloth at all. The chapter therefore has no Ring 1 museum objects yet, and that is recorded honestly rather than papered over.

ID	Node	What it is	Source and status	Role
PL-020	Madjedbebe (place)	Rock shelter, Mirarr Country, Northern Territory	Excavation reports; both dates shown	Deep time; the contested line
PL-021	Budj Bim (place)	Gunditjmara aquaculture landscape	UNESCO; Gunditjmara-authored material to be sought first	Engineering without metal

ID	Node	What it is	Source and status	Role
WS-030	Ground-edge axe fragment, Carpenter's Gap	44,000–49,000 years old	Published finds; image rights to request	The earliest known ground edge
WS-031	<i>Rebbelib</i> navigation chart	19th–early 20th century, Marshall Islands	Met 1978.412.826, not public domain; Smithsonian NMNH holds 15 charts, rights to check	The model of the sea
WS-032	Lapita dentate-stamped sherd	About 3,300 years ago	Excavated material; museum image to be licensed	The pottery that maps the voyages
WS-033	<i>Masi</i> (barkcloth), Lau Islands, Fiji	Late 19th–early 20th century	Met 1977.395.3, not public domain	Case only, until a licensed example is found (Te Papa and Australian collections next)
PR-010	Wayfinding (practice)	Hawaiian and Micronesian non-instrument navigation	Polynesian Voyaging Society, in its own words	The living practice

Living practice. Wayfinding, told by the people who revived it: Mau Piailug's role and the 1976 and 1980 voyages, from the Polynesian Voyaging Society's own history. Budj Bim's eel farming is the second, with the 2007 native-title recognition as its recent chapter.

Protocol. This chapter runs on the strictest settings. Aboriginal, Torres Strait Islander, Māori and Pacific material follows CARE and, for Australia, Creative Australia's protocols: community-authored or community-approved sources first; no sacred, secret or funerary material; named Country and community for every node; consultation flagged on every Apply. Human remains and ancestral objects are never shown. The Historian pre-reads as a Pacific specialist, and this chapter goes to a paid sensitivity read as soon as budget exists (P11).

The walk

1. **Guess.** When did people first make things in Australia? A slider from 5,000 to 100,000 years.
2. **The axe.** A ground edge 44,000–49,000 years old; grinding as a technology, and a first that is "earliest known".
3. **The date argument.** Madjedbebe: 65,000 or about 50,000, both readings drawn.
4. **Farming the water.** Budj Bim: channels and weirs in volcanic stone, six thousand years of eel harvest, a system still running.
5. **Pots that crossed water.** Lapita: one decorated ware from the Bismarcks to Fiji in a few centuries, the trail of a voyaging people.
6. **The model of the sea.** A *rebbelib*: swells, currents, islands, memorised and left on land.
7. **The revival.** Hōkūle'a, 1976 and 1980: a navigator from Satawal teaches Hawai'i to sail home.
8. **Who may use this.** Australia's protocols, and what a designer must do before drawing on Country.

Apply

Lane	Task
Both	A reference note for one Pacific or Australian practice, with the community named, the source community-authored, and the consultation status filled in
Maker	One technique in your world that works without writing, and how you would document it with its holders
Briefer	Rewrite a brief line that says "tribal" or "Aboriginal-inspired" into one that names Country, people and consent

How we will know it works. Every Apply names a community and a consultation status. At least 80% of learners can state both Madjedbebe readings. No learner submission uses "primitive", "tribal" or "discovered" for these peoples (writing check).

Links out. F1.5 (grinding as an early technology), F1.13 (Lapita and the ocean), F1.25 (Practice Reading), F1.26 (protocols; this chapter is the protocol unit's live case), F1.28a (Hōkūle'a as reconstruction), F1.29 (oral transmission), F1.31 (the regional slot's model of

local authorship).

What changed. Australia joins the chapter (Q1). Navigation stays the device and gains grinding and aquaculture as deep-time anchors. The unit currently has zero licensed Ring 1 museum objects; the harvest must look to Te Papa, Australian state collections and the Smithsonian, and to community publishing. The Met search filter bug is added to the pipeline findings.

F1.10 The Islamic world as a network, read through routes and the waqf

The chapter’s two devices hold. Routes explain how paper, cobalt, cotton and coins moved; the waqf explains who paid for the buildings, lamps and tiles, and why endowment deeds are the best record of named craftsmen the pre-modern world has. The deep dive also found the chapter’s best copying case: three “Mamluk” glass pieces in Cleveland are 19th-century Paris and European work, one of them inscribed with a Mamluk sultan’s name.

Reach. From Baghdad’s first paper mill (790s) to the Ottoman 16th century, across Egypt, Syria, Iran, Anatolia, al-Andalus, the Swahili coast and the Indian Ocean, with the Gulf as the hook for the regional vertical. **Groupings used:** a belief tradition (Islam, with non-Muslim makers inside it named); polities (Abbasid, Fatimid, Mamluk, Ottoman); a network (the Indian Ocean; the Cairo–Hormuz–India route); an institution (the waqf); maker communities (calligraphers; Iznik potters; Mamluk builders); a diaspora (the Geniza’s Jewish traders).

Claims

Claim	Confidence	Source
The first paper mill in Islamic lands was set up in Baghdad in 794–95; paper then spread to Syria, Iran, Central Asia, Egypt and North Africa, and to Europe through Spain and Sicily	Documented	Bloom, <i>Paper Before Print</i> , review

Claim	Confidence	Source
Mamluk endowment deeds (13th–16th centuries) name the craftsmen appointed to maintain endowed buildings, set their duties and wages, and were enforced through the judge's court	Documented	Abdulkarim, "Building Practices in Light of Mamluk Endowment Documents"; her thesis
Abraham Ben Yiju, a Jewish trader from Tunisia, ran a bronze workshop in India in the 1130s and 1140s; its accounts survive in the Cairo Geniza	Documented	Princeton Geniza Project, CUL Or.1080 J95 ; Goitein and Friedman, <i>India Traders</i> (2008)
Kilwa's merchants traded across the Indian Ocean from the 13th to the 16th century; the Great Mosque's vaults were set with Chinese porcelain	Documented	UNESCO (shared with F1.13)
The cobalt in early Islamic glazed ceramics matches the cobalt later shipped to Jingdezhen, most likely from Anarak in Iran	Probable	PNAS 2020 (shared with F1.13)
Arabic calligraphy was inscribed by UNESCO in 2021 at the request of sixteen states; it is taught through schools and apprenticeship, and practised from marble carving to spray-painted walls	Documented	UNESCO
Gulf lenj boat-building and navigation are on UNESCO's urgent safeguarding list (2011)	Documented	UNESCO
Three enamelled glass pieces in Cleveland catalogued as Mamluk-style are European work of the late 1800s; one carries an inscription naming the sultan Muhammad ibn Qala'un	Documented	Cleveland records 1981.10, 1944.235 and 1944.488, read live

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-021	Footed bowl, probably Kashan	Late 1100s–early 1200s	Cleveland 1915.581, CC0	Bought 1915	Cobalt in Iran (shared with F1.13)
WS-040	Mosque lamp, Iznik	1585–95	Cleveland 1944.236, CC0	A New York dealer, 1944	A fitting made for an endowed building
WS-041	Large dish with artichokes, Iznik	c. 1535–40	Cleveland 1995.17, CC0	Named collections, then a London dealer, 1995	The Iznik workshop at its peak
WS-042	Tile spandrel with floral sprays, Iznik	c. 1570–75	Cleveland 2004.70, CC0	A London dealer, 2004; earlier history absent. Flagged	Architectural tile; provenance to confirm
WS-043	Cover with four trees of life, Egypt	4th–5th century	Met 90.5.902, public domain	Given 1890; provenance unchecked	Egypt's weaving before Islam: the chapter opens on continuity
WS-006	Block-printed cotton, found in Egypt	1200s–1300s	Cleveland 1929.845, CC0	Acquired 1929	Indian cotton in Mamluk Egypt
WS-044	Bottle, enamelled glass, Europe	c. 1900	Cleveland 1944.488, CC0	A New York dealer, 1944	The copying case: a Mamluk sultan's name on a European bottle
WS-045	Mosque lamp by Philippe-Joseph Brocard, Paris	Late 1800s	Cleveland 1981.10, CC0	Brocard, then dealers	The same, with a named maker
DOC-001	Ben Yiju's workshop accounts	1132–1149	Cambridge University Library Or.1080 J95; image rights to check	—	A workshop's own ledger

Living practice. Arabic calligraphy, from apprenticeship to calligraffiti; and, for the Gulf, lenj building. Both from UNESCO, with practitioners' own accounts to be added.

Protocol. Qur'an folios are shown only from institutions that publish them, with no cropping or overlay on the text. The chapter names non-Muslim makers within the network (the Geniza traders, Coptic weavers) so "Islamic" describes the network, not a test of who belongs.

The walk

1. **Guess.** Who paid for a mosque lamp, and who kept it lit?
2. **Before.** A 4th-century Egyptian weaving: the makers who were here first, and stayed.
3. **Paper.** Baghdad's mill in the 790s and the routes paper took, with the Talas legend marked as legend.
4. **The deed.** A Mamluk endowment: the craftsmen it names, the wages it sets, the judge who enforced it. The waqf as a commissioning regime (to F1.17).
5. **The lamp.** Iznik, 1585–95: a fitting made to hang in an endowed building.
6. **A ledger from India.** Ben Yiju's bronze workshop, 1130s, in a Jewish trader's hand: the network's own record.
7. **Blue.** Kashan's bowl and the cobalt that went east.
8. **The copy.** Brocard's Paris lamp and a European bottle with a sultan's name: Europe's 19th-century "Mamluk" revival (to F1.14).
9. **The hand today.** Calligraphy from the reed pen to the spray can; the Gulf's boatbuilders.

Apply

Lane	Task
Both	Read one endowment deed extract and list who it names, what they were paid, and what they had to maintain
Maker	Name a route your world could use and what travels it: goods, materials, skills or people
Briefer	Draft the maintenance clause a brief should carry, on the waqf's model: who keeps the thing working after handover

How we will know it works. At least 80% of learners can name the waqf as a commissioning regime and give one thing a deed records. At least 70% tell the 19th-century revival glass from Mamluk work when shown both. Briefer maintenance clauses name a responsible party.

Links out. F1.5 (paper), F1.13 (cobalt, Kilwa, the Gulf), F1.14 (the Brocard copy), F1.15 (the Grammar of Ornament's Islamic plates), F1.17 (the waqf), F1.28 (the Geniza as a primary source), F1.31 (the Gulf hook), the new West Asia chapter (what came before).

What changed. The chapter opens on pre-Islamic Egypt to show continuity, gains the Geniza as a primary-source case, and adds the revival-glass copying case. Calligraphy joins lenj building as living practice. One Iznik record is flagged for provenance.

F1.7 The Americas, read through fibre

Fibre is the right device because in the Andes it came first: cotton before pottery, indigo before anyone else, and a state that ran its accounts on knotted cords. The deep dive widened the chapter north and east, to Diné weaving after the Long Walk and to the Amazon’s mound and garden cities, and it found a clean repatriation case in Paracas.

Reach. The Andes, the Amazon, Mesoamerica and the North American Southwest, from 7,800 years ago to the present; the Arctic, Northwest Coast and Eastern Woodlands stay logged for the world hub. **Groupings used:** horizons (Early, Middle and Late in the Andes); archaeological cultures (Paracas, Marajoara); polities (the Inka state); a community (Diné); a maker community (Diné weavers, named as women artists in the records); the environment lens (desert coast, highland, rainforest).

Claims

Claim	Confidence	Source
Cultivated cotton is at least 7,800 years old in the Andes; indigo-dyed cotton from Huaca Prieta, about 6,200 years old, is the earliest known indigo anywhere	Documented	Splitstoser et al. 2016 (shared with F1.5)
Caral's monumental centre and irrigated fields date to 2627–1977 BCE, before pottery in the region	Documented	Shady, Haas and Creamer 2001
Khipus recorded census and tribute; the six Santa Valley khipus match a 1670 revisit document tributary by tributary, and may have signed names by colour	Documented for counts; interpretive for names	Medrano and Urton 2018, <i>Ethnohistory</i>

Claim	Confidence	Source
The Open Khipu Repository holds open data on surviving khipus; the system remains undeciphered	Documented	Open Khipu Repository
Road-linked garden cities stood in the Upano Valley from about 500 BCE to 300–600 CE	Documented	Rostain et al. 2024
The Marajoara built mound settlements with a funerary ceramic tradition on Marajó Island at the Amazon's mouth	Documented	Roosevelt, <i>Moundbuilders of the Amazon</i> (open copy); dates to confirm from the monograph
Paracas textiles left Peru illegally after their discovery; Gothenburg's collection was shown in 2008 as "A Stolen World" and its ownership returned to Peru, the first four textiles in 2014	Documented	Javér, conservation paper
Churro sheep, brought by the Spanish in 1598, became the base of Diné weaving; in 1863–64 the US Army destroyed flocks and forced more than 8,000 Diné on the Long Walk to Bosque Redondo; they returned in 1868	Documented	Hubbell Trading Post, National Park Service
Diné weavers used Germantown mill-spun yarns and bayeta (unravelling trade cloth) in the late 1800s, and trading posts steered weaving from wearing blankets to rugs	Documented	Cleveland records 1921.556 and 1937.903; NPS

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-050	Mantle, Paracas (Cavernas style)	c. 300 BCE–200 CE	Cleveland 1946.226, CC0	Given by the Norwebs in 1946; before that unknown. Passes the 1970 test; its removal from Peru is undocumented	Embroidery on camelid fibre; taught with the Gothenburg case
WS-051	Tunic, Paracas	c. 300 BCE–200 CE	Cleveland 1946.227, CC0	Same	Warp substitution and embroidery
WS-052	Turban band (<i>llauro</i>), Paracas	c. 300–100 BCE	Cleveland 1946.228, CC0	Same	Needle netting
WS-053	Inka khipu	c. 1400–1532	Cleveland 1940.469, CC0	No provenance in the record. Flagged	The record without writing
WS-007	Wari tunic	650–1000	Met 2021.146, public domain	Given 2021; provenance unchecked	The Middle Horizon (shared with F1.5)
WS-054	Wearing blanket, Diné, woman artist	c. 1875–80	Cleveland 1921.556, CC0	No provenance in the record	Handspun, Germantown and bayeta in one cloth: the trade in the weave
WS-055	Fourth-phase chief blanket style rug, Diné	c. 1900	Cleveland 1937.903, CC0	Given by Amelia Elizabeth White, 1937	From blanket to rug: the trading post's market
WS-004	Chavín-style gold plaque	c. 500–200 BCE	Cleveland 1938.431, CC0	Owned by 1938	The Early Horizon (shared with F1.5)
WS-022	Basin, Puebla	c. 1650	Met 12.3.1, public domain	Given 1912	Mexico after contact (shared with F1.13)
DAT-001	Open Khipu Repository	—	MIT-licensed data	—	Ring 2 dataset

Living practice. Diné weaving, with the Navajo-Churro sheep restoration since the 1970s, told from Diné sources first (the Diné weavers' own organisations and the NPS material as a second voice). Andean weaving cooperatives are the second candidate, to be sourced from the weavers' own publications.

Protocol. Paracas material comes from burials, so the bundles and bodies are never shown; the textiles are shown as textiles, with their looting history stated. Diné objects follow CARE: no ceremonial textiles, and the museum's "woman artist" attribution is kept as the museum's wording beside our reading. Mesoamerican codices appear only as facsimiles from the holding libraries.

The walk

1. **Guess.** Which came first in the Andes: cotton, pottery or cities?
2. **Blue before anyone.** Huaca Prieta's indigo cotton, 6,200 years old.
3. **A city without pots.** Caral: irrigation and monuments before ceramics.
4. **Embroidered for the dead.** The Paracas mantle, and "A Stolen World": the Gothenburg return, and Cleveland's own record's silence about Peru.
5. **The knotted record.** The khipu; what the Santa Valley khipus proved and what stays undeciphered.
6. **The horizon.** The Wari tunic: one style across the Andes, a spread of style rather than an empire.
7. **Under the forest.** Upano's garden cities and Marajó's mounds: the Amazon as a place of making.
8. **Sheep, war, and yarn.** Churro sheep, the Long Walk, Germantown yarn and bayeta in one Diné blanket.
9. **The rug.** The trading post turns a wearing blanket into a floor covering for a distant market.

Apply

Lane	Task
Both	Read one textile's technique layer: fibre, structure, dye, and where each came from
Maker	One fibre or material in your world whose supply was changed by force or by market, and what your design owes to that
Briefer	Write the provenance line a brief must require for any "pre-Columbian-inspired" reference

How we will know it works. At least 80% of learners can name one Andean recurrence that preceded pottery. At least 70% describe the khipu as a record without calling it writing or dismissing it. Briefers' provenance lines name a country of origin and an export status.

Links out. F1.5 (cotton, Caral, Upano), F1.13 (the Puebla basin), F1.18 (women's making: Paracas and Diné weavers), F1.19a (war and making: the Long Walk), F1.22a (use: blanket to rug), F1.24 (the Gothenburg return), F1.28 (the Geniza's counterpart, the khipu, as a source), the steppe chapter (fibre economies compared).

What changed. The chapter now spans the hemisphere: Amazonia and the Southwest are added; three Paracas records are verified; the khipu is flagged for provenance; the Diné blanket becomes the chapter's case of a material chain remade by conquest and market.

F1.8 East Asia, read through the workshop

The workshop is the device because East Asia's great objects were made by organised groups working to written standards: bronze foundries casting in piece-moulds, a state building code with a modular unit, an imperial porcelain bureau ordering hundreds of thousands of pieces, and a city that still has 100,000 ceramic workers. The deep dive's main finding is that the living practice is not a revival but a working industrial city with a young migrant workforce.

Reach. China, Korea and Japan from the Shang (c. 1600 BCE) to today; the Jōmon deep end sits in F1.5. Mongolia and the Tibetan plateau go to the steppe chapter and the world hub. **Groupings used:** dynasties (Shang, Song, Goryeo, Yuan, Ming); maker communities (foundries; the Jingdezhen bureau and the civil kilns; the *jingpiao* newcomers); a movement (Mingei, as the Atlas Logic example); a technique family (piece-mould casting, celadon inlay, underglaze blue).

Claims

Claim	Confidence	Source
The <i>Yingzao Fashi</i> (State Building Standards) was compiled by Li Jie, a state building superintendent, in 1100 and published in 1103; its 34 chapters set units, design standards, labour estimates and material data	Documented	Guo, Architectural History
Song Yingxing wrote in 1637 that a piece of clay passes through 72 pairs of hands before it becomes porcelain	Documented	<i>Tiangong Kaiwu</i> (1637), quoted in a Jingdezhen survey
The Ming court's 1433 order of 443,500 pieces pushed production into the civil kilns under imperial supervision ("moulded by the imperial kiln, fired by the civil kilns")	Documented, from a secondary summary citing UNESCO	Same survey; to replace with the UNESCO tentative-list text
Early blue-and-white used cobalt imported from Iran	Documented	PNAS 2020 (shared with F1.13)
Oracle-bone inscriptions of Wu Ding's reign are radiocarbon-dated 1254–1197 BCE	Documented	Liu et al. 2020 (shared with F1.5)
Printed dharani for the One Million Pagodas, 764–770, and Jikji, 1377, are landmarks of printing	Documented	Cambridge Digital Library; UNESCO (shared with F1.5)

Claim	Confidence	Source
Around 2000, ten state ceramic factories in Jingdezhen closed; about 100,000 ceramic workers remained; since 2016 the Taoxichuan district selects 1,200 stallholders a month from over 4,000 applicants; more than 100,000 young people have moved to the city in a decade	Documented, from state media	People's Daily, July 2025 ; to be balanced with an independent account
Mingei was named by Yanagi Sōetsu in the mid-1920s; the Japan Folk Crafts Museum opened in 1936; historians read the movement as entangled with Japan's imperial policy	Documented; the reading is one of several	SamuraiWiki with references ; review of Brandt

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-002	Wine vessel (<i>jue</i>), inscribed	c. 1200 BCE, Shang	Cleveland 1960.42, CC0	Acquired 1960	Piece-mould casting; the clan inscription (shared with F1.5)
WS-060	Tripod cauldron (<i>ding</i>)	1200–1100 BCE, Shang	Cleveland 1960.288, CC0	Acquired 1960; earlier history not in the record. Flagged	The foundry's scale
WS-061	Jar with cover, Jun ware	1100s, Northern Song or Jin	Cleveland 1957.36, CC0	A seller's number only. Flagged	A glaze that runs and pools: the kiln as co-maker
WS-062	Vase with inlaid lotus, plum and bamboo	1300s, Goryeo	Cleveland 1918.471, CC0	Bought by Langdon Warner, given 1918	Korea's inlay technique (<i>sanggam</i>)

ID	Object	Date	Holder and licence	Provenance	Role
WS-063	Bowl with inlaid waterfowl, willow and reed	1300s, Goryeo	Cleveland 1918.458, CC0	Bought by Warner in Korea; given 1918	The same; Warner's collecting as a case for F1.24
WS-020	Jar with lion-head handles, Jingdezhen	1300s, Yuan	Cleveland 1962.154, CC0	Deep dive F1.13	Blue-and-white for export
WS-008	One of the One Million Pagodas	764–770	Met 30.47a–c, public domain	Given 1930	Printing as mass production
WS-005	Fire-flame vessel, Jōmon	c. 2500 BCE	Cleveland 1984.68, CC0	Flagged (Deep Dives 1)	The deep end (shared with F1.5)
DOC-002	<i>Yingzao Fashi</i> plates	1103 (later editions)	Public domain; Wikisource scans	—	The modular code

Living practice. Jingdezhen as a working city: 100,000 ceramic workers, a training and stall system, and young makers selling by livestream. The state-media source is clear about its purpose, so the unit pairs it with a maker's own account (an independent interview or documentary, to be sourced) and reads both.

Protocol. Nothing sacred or funerary is shown in the taught set; Shang ritual bronzes are shown as foundry products with their ritual use named. The Korean pieces carry Warner's collecting history as a case, since Korea under Japanese rule (1910–45) is where much Goryeo celadon left the country.

The walk

1. **Guess.** How many people does it take to make one porcelain cup?
2. **The foundry.** The Shang *jue* and *ding*: piece-moulds, alloys and a clan name cast into the metal.
3. **The code.** The *Yingzao Fashi*, 1103: a state manual with standard units, labour estimates and paint recipes.
4. **The kiln decides.** Jun ware: a glaze the potter cannot fully control, and a workshop that learned to want that.
5. **Inlay.** Goryeo celadon: cut, filled, glazed; and how these pieces reached Cleveland in 1918.
6. **Seventy-two hands.** Jingdezhen, 1637: Song Yingxing's count; the imperial bureau's 443,500-piece order and the civil kilns that filled it.

7. **Blue from Iran.** The Yuan jar and the cobalt route (to F1.13).
8. **A million copies.** The pagoda and its printed invocation: mass production in the 760s.
9. **The city now.** Jingdezhen's newcomers: 100,000 in a decade, stalls, incubators and livestreams; Mingei as the counter-story of the handmade (to the grouping page).

Apply

Lane	Task
Both	Draw the operational sequence of one object in your world as a chain of hands: who does each step, and how many there are
Maker	Write the one rule of your practice you would put in a <i>Yingzao Fashi</i> of your own
Briefer	Specify the workshop model a brief assumes (studio, factory, distributed makers) and what that means for credit

How we will know it works. At least 80% of learners can name the workshop, not the individual, as the maker of at least two objects. At least 70% draw a sequence of five or more hands. Briefers' credit clauses name a group, not only a brand.

Links out. F1.5 (writing, pottery, printing), F1.13 (cobalt), F1.14 (blue-and-white copied in Iznik, Delft and Puebla), F1.17 (the imperial bureau as commissioner), F1.19 (industry before the machine), F1.20 (Mingei), F1.23 (credit to a workshop), F1.24 (Warner in Korea), F1.28a (the *Yingzao Fashi* as a text to build from).

What changed. The living practice becomes the working city rather than a revival, with a source-balance rule. Two Cleveland records are flagged for provenance. Mingei moves to the grouping page and F1.20, so the chapter's spine stays on the workshop.

F1.9 South and Southeast Asia, read through cotton

Cotton is the device because India's cloth was the world's first mass export: printed in Gujarat, worn in Cairo and Java, banned in London, and undone by the machine. The deep dive found the chapter's best primary evidence (dated Gujarati fragments dug up in Egypt), its sharpest law (Britain's 1721 ban on wearing printed calico), and a live restitution case for the temple bronze.

Reach. The Indus to Bengal, Tamil Nadu, Gujarat, and island Southeast Asia, from 2500 BCE to today; mainland Southeast Asia (Đông Sơn, Khmer) and the Himalaya are logged for the world hub. **Groupings used:** polities (Chola, Mughal, the East India Companies); maker communities (Gujarati printers, Dhaka weavers, Chola bronze-casters); a network (the western Indian Ocean; the *patola* trade to Indonesia); a belief tradition (the temple as commissioner); materials (cotton, indigo).

Claims

Claim	Confidence	Source
Gujarati block-printed and painted cottons found in Egyptian deposits are the earliest material traces of the medieval western Indian Ocean trade; they date from the Fatimid to the early Mamluk periods (10th to early 16th centuries) and similar cloths were traded as far as Indonesia	Documented	Barnes, <i>Textile History</i> 2017 ; Barnes, <i>Indian Block-Printed Textiles in Egypt</i> (1997)
Bengal's handloom textiles held a leading place in Asian and North African markets long before European companies, and were a staple of the companies' export trade until machine-made cloth arrived in the early 19th century	Documented	Chaudhury, <i>Modern Asian Studies</i>
In 1721 Parliament prohibited the use and wear of printed, painted, stained or dyed calicoes in clothing and furnishing, to protect wool and silk	Documented	The Act, 7 Geo. I c. 7, Wellcome Collection
Two Old World cottons were domesticated separately; India's <i>Gossypium arboreum</i> is one	Documented	Renny-Byfield et al. 2016 (shared with F1.5)

Claim	Confidence	Source
Jamdani, a sheer patterned cotton woven on handlooms around Dhaka with a discontinuous-weft technique, was inscribed by UNESCO in 2013; it is taught mainly by parents to children in home workshops	Documented	UNESCO
The National Gallery of Australia bought a Chola Nataraja for US\$5 million in 2008 from Subhash Kapoor; police in Tamil Nadu recorded the theft of a Nataraja from Sripuranthan in 2008; Australia returned the bronze to India in September 2014	Documented	NGA statement
Abraham Ben Yiju ran a bronze workshop in India in the 1130s and 1140s	Documented	Geniza (shared with F1.10)
Whether the Indus signs are writing is argued	Contested	Rao 2009 against Farmer, Sproat and Witzel 2004 (shared with F1.5)

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-006	Block-printed cotton fragment, found in Egypt	1200s–1300s	Cleveland 1929.845, CC0	Acquired 1929	Catalogued as “Egypt, Mamluk”; Barnes’s work suggests the Gujarat question must be asked. The card shows both
WS-070	Nataraja, Shiva as Lord of Dance	1000s, Chola	Cleveland 1930.331, CC0	C. T. Loo, New York, sold 1930	Lost-wax casting for a temple; the NGA case beside it
WS-071	Shiva and Parvati	c. 1020–40	Cleveland 1961.94, CC0	A New York dealer, 1961	The same workshop tradition
WS-072	Jamdani sari	Living practice	Weavers’ own images, to license	—	The living cloth

ID	Object	Date	Holder and licence	Provenance	Role
WS-073	<i>Patola</i> double-ikat silk, Patan	To source	Not in Cleveland's CC0 set; Met or Rijksmuseum to check	—	Gujarat's silk for Indonesia
WS-074	Indus seal	c. 2500 BCE	Met or Cleveland to check at record level	—	The signs (shared with F1.5)
DOC-003	Dutch Company order for Bengal cloth, 1701	—	Cited in Chaudhury; archive at the Nationaal Archief	—	Cloth ordered by the thousand

Cleveland's CC0 set has no chintz, batik or ikat, so the textile core of this chapter still needs its objects from the Rijksmuseum (which publishes chintz and Indonesian textiles openly) and from the Met at record level. That is the first harvest task for this unit.

Living practice. Jamdani weaving around Dhaka, from UNESCO and from weavers' cooperatives. Kalamkari and block printing in Gujarat and Andhra are the second candidates.

Protocol. Temple bronzes are images in worship; the Cleveland pieces are shown as cast objects with their function named, and the Sripuranthan theft is told with the police record, not as a museum mystery. Indonesian textiles with ritual use follow CARE.

The walk

1. **Guess.** Where was the cloth in a Cairo household of 1300 made?
2. **Dug up in Egypt.** Gujarati printed cottons in Fustat: the cloth trade's earliest surviving pieces, and how Cleveland's fragment is labelled.
3. **The workshop by the sea.** Ben Yiju's bronze accounts from India (the Geniza), and the Indian Ocean as one market.
4. **Cast for a temple.** The Chola Nataraja: lost-wax casting, the temple as commissioner, and the 2014 return from Canberra.
5. **Ordered by the thousand.** Dutch and English orders for Bengal cloth in the 1700s; muslin so fine it needed its own names.
6. **Banned.** 1721: it becomes illegal in Britain to wear printed calico. What protection did, and to whom.
7. **Undone.** Machine cloth after 1800 and the end of Bengal's export trade.
8. **Still woven.** Jamdani around Dhaka: the discontinuous weft, learned at home.

Apply

Lane	Task
Both	Trace one textile in your world from fibre to wearer, naming every hand and every border it crossed
Maker	One technique of yours that a law or tariff could stop, and what you would do
Briefer	Write the origin line a brief must require for any "Indian-inspired" print, and the check against it

How we will know it works. At least 80% of learners can explain why a Gujarati cloth is catalogued as Egyptian. At least 70% state what the 1721 act banned and why. Briefers' origin lines name a region, a technique and a maker community.

Links out. F1.5 (cotton, the Indus signs), F1.10 (the Geniza; cotton in Mamluk Egypt), F1.13 (the Indian Ocean), F1.14 (chintz copied in Europe), F1.17 (the temple and the Company as commissioners), F1.19 (the machine), F1.24 (the Nataraja's return), F1.18 (Dhaka's weavers).

What changed. The chapter leads with Egypt's Gujarati cottons and adds the 1721 Act and the Nataraja return. Its textile objects are not yet licensed; the Rijksmuseum is added as a harvest source.

F1.11 Europe and the Mediterranean, read through the guild

Europe is one world among several, and the guild is its device: a written rule for who may make, how they learn, and what they may sell. The deep dive widened the chapter to Mediterranean antiquity (Q1), which turns out to strengthen the device, because Roman potters stamped their names on mass-produced bowls a thousand years before Paris wrote its guild statutes. It also found a living guild: French *compagnonnage*, with 45,000 members and a masterwork still required.

Reach. The Mediterranean and Europe from Attic pottery (6th century BCE) to the Great Exhibition (1851), with Byzantium, the Balkans, Scandinavia and the Sámi logged for the world hub. **Groupings used:** polities (Athens, Rome, Paris, the Dutch Republic, Britain); maker communities (Attic workshops, Arretine firms, Paris trades, Delft potteries, Wedgwood's factory, the *compagnons*); a style (black-figure); a movement (the reform circle around the 1851 exhibition); networks (the Roman Mediterranean; the Atlantic, handed to F1.13).

Claims

Claim	Confidence	Source
Roman red-gloss pottery (<i>terra sigillata</i>) was mass-produced in several parts of the empire and exported to all of it; its stamps name the people involved in production, including slaves and freedmen, and the largest Arretine firms had dozens of potters	Documented	Fülle, <i>Journal of Roman Studies</i>
The <i>Livre des métiers</i> , compiled under Étienne Boileau in 13th-century Paris, records the statutes of the city's trades	Documented	The 1837 edition, Internet Archive (the count of trades to be taken from the edition)
Chinese blue-and-white was imitated in Delft on tin-glazed earthenware, and Delft was imitated in England	Documented	Cleveland records 1969.254 and 1986.31, read live; the chain to F1.14
In 1721 Britain banned the wearing of printed calico to protect wool and silk	Documented	The Act, Wellcome Collection (shared with F1.9)
Wedgwood's jasperware copy of the Portland Vase, c. 1790, reproduced an ancient Roman cameo glass in a new ceramic body	Documented	Cleveland 1918.823, read live
The Great Exhibition of 1851 showed more than 100,000 objects to six million visitors in a purpose-built iron and glass building; Owen Jones, one of its superintendents of works, published <i>The Grammar of Ornament</i> in 1856 with 100 plates in 19 categories	Documented	National Museums Scotland

Claim	Confidence	Source
French <i>compagnonnage</i> was inscribed by UNESCO in 2010: about 45,000 people, training of about five years on the move between towns, and a masterwork judged by the <i>compagnons</i> before one may teach	Documented	UNESCO
Copper was smelted at Belovode, Serbia, about 5000 BCE	Documented	Radivojević et al. 2010 (shared with F1.5)

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-080	Black-figure amphora, Attic	c. 530–520 BCE	Cleveland 1927.433, CC0	Bateman collection, London; Florence dealers; sold 1927	The potter's and painter's workshop; the Mediterranean before Europe
WS-081	Black-figure neck-amphora, Attic	515–510 BCE	Cleveland 1970.16, CC0	Only “1970—” in the record. Flagged	Same
WS-082	<i>Terra sigillata</i> bowl with a maker's stamp	1st century	To source from the Met or the British Museum at record level	—	The stamped mass product
WS-083	Plate, Delft	1700s	Cleveland 1969.254, CC0	No provenance in the record	The copy of a copy (to F1.14)
WS-084	Plate, English delftware	Mid-1700s	Cleveland 1986.31, CC0	A Cleveland auction	Same, one step on
WS-085	Portland Vase, Wedgwood jasperware	c. 1790	Cleveland 1918.823, CC0	In the museum since 1918	The factory reproduces antiquity
AS-001	<i>The Grammar of Ornament</i> , plates	1856	Public domain	—	The exhibition as taxonomy (to F1.15)
DOC-004	<i>Livre des métiers</i>	13th century, 1837 edition	Public domain, Internet Archive	—	The guild rule in the guild's words

ID	Object	Date	Holder and licence	Provenance	Role
WS-010	Benin plaque	1500s–1600s	Cleveland 1999.1, CC0	Deep Dives 1	Rhineland brass: Europe's metal in Africa's casting (to F1.6)

Living practice. *Compagnonnage*: the *tour de France*, the masterwork, and the initiation rites, from UNESCO and from the *compagnons'* own houses.

Protocol. Antiquities follow the 1970 test; the 1970 amphora is flagged. Enslaved and freed makers named on Roman stamps are named as such in the unit text.

The walk

1. **Guess.** Who was allowed to make a chair in Paris in 1268?
2. **Signed in Athens.** The amphora: a workshop of potter and painter, and the first named makers in this world.
3. **Stamped in Arezzo.** *Terra sigillata*: mass production across an empire, with slaves' and freedmen's names in the clay.
4. **The rule book.** The *Livre des métiers*: apprenticeship, hours, who may sell, in the trades' own statutes.
5. **Copper before all this.** Belovode, 5000 BCE: the Balkans in Europe's story.
6. **Copying China.** Delft copies Jingdezhen, England copies Delft; Britain bans Indian calico (to F1.14 and F1.9).
7. **The factory.** Wedgwood's Portland Vase: antiquity reproduced by division of labour, and a brand name.
8. **The exhibition.** 1851: six million visitors, 100,000 objects, and a grammar written to sort the world's ornament (to F1.15).
9. **Still on the road.** *Compagnonnage* today: five years, many towns, one masterwork.

Apply

Lane	Task
Both	Write the three rules a guild for your practice would set: entry, training, and what may be sold
Maker	Name the masterwork you would have to make to be admitted, and what it would prove

Lane	Task
Briefer	Draft the qualification clause a brief should carry: what the maker must have done before, and how it is shown

How we will know it works. At least 80% of learners place mass production before the machine (Arezzo) and Europe after other worlds in at least one technique (copper, blue-and-white). At least 70% can state one guild rule and whom it protected. Briefer qualification clauses ask for evidence, not titles.

Links out. F1.5 (Belovode), F1.6 (brass), F1.9 (calico), F1.13 (the Atlantic), F1.14 (Delft, Wedgwood), F1.15 (Jones), F1.17 (the guild as a commissioning regime; the exhibition as patron), F1.19 (industry), F1.23 (stamps as credit), F1.26 (guild rules as the first protocol).

What changed. The chapter reaches back to Athens, Rome and the Balkans (Q1). The guild device now runs from Roman stamps to *compagnonnage*. One amphora is flagged for provenance; the *terra sigillata* exemplar is still to be licensed.

New chapter: West Asia before Islam, read through the account

The proposed device is the account: the world's earliest writing is a list of rations, its earliest known glass-blowers signed their cups, and its earliest copper trade is known from bills of lading. West Asia is where making first became a matter of record, so the chapter teaches record-keeping as a technology alongside metal, glass and cloth. It also puts the Gulf into F1's first millennia, which the regional vertical will need.

Reach. Mesopotamia, Iran, the Levant, Anatolia and the Caucasus, Arabia and the Gulf, from the first cities (4th millennium BCE) to the eve of Islam. **Groupings used:** polities (Uruk, the Assyrian and Achaemenid empires, Rome in the east); archaeological cultures (Luristan bronzes as a style-and-culture label); a network (the Gulf copper route: Magan to Sumer); maker communities (scribes; Levantine glass-blowers; Luristan smiths); materials (copper, glass, wool, purple dye).

Claims

Claim	Confidence	Source
The earliest cuneiform tablets are administrative accounts, such as one recording the distribution of barley and emmer, about 3100–2900 BCE, probably from Uruk	Documented	Met 1988.433.2, read live
Writing may have been invented three or four times; Mesopotamia's is the earliest securely dated	Contested (the count)	CL-001 (shared with F1.5)
Glass-blowing was invented towards the end of the first century BCE in the coastal Levant; the earliest evidence is from Jerusalem; some of the only named glass-workers of the period, such as Ennion, were Greek-speaking Syrians	Documented	Derrick, <i>The Conversation</i>, 2026 ; to be backed by the excavation report
Magan, the copper land of Sumerian texts, is identified with Oman, where copper was mined and smelted from the Early Bronze Age (Hafit and Umm an-Nar periods, c. 3200–2000 BCE)	Documented	Weisgerber and others, Magan: The Land of Copper
Copper was smelted in the Balkans about 5000 BCE, earlier than any secure Near Eastern evidence, which challenges the single-origin model	Documented (the finding); contested (what it means)	Radivojević et al. 2010 (shared with F1.5)

Claim	Confidence	Source
Luristan bronzes (c. 800–600 BCE) are cast horse-gear and fittings from western Iran, mostly looted rather than excavated	Documented for the objects; the looting history to source	Cleveland 1980.102 and 1961.33, read live
Egypt's earliest writing, c. 3320 BCE, developed in parallel with Mesopotamia's; whether it borrowed the idea is argued	Contested	Kahl 2001 (shared with F1.5)

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-090	Cuneiform tablet: distribution of barley and emmer	c. 3100–2900 BCE	Met 1988.433.2, public domain	Bought 1988 (Sackler gift); earlier history not in the open data. Flagged, like its sibling WS-001	The account as the first text
WS-001	Proto-cuneiform tablet with seal impressions	c. 3100–2900 BCE	Met 1988.433.1	Same purchase. Flagged	F1.1's anchor; now with a documented sibling
WS-091	Horse bit with winged ibex cheekpieces, Luristan	800–600 BCE	Cleveland 1980.102, CC0	No provenance in the record. Flagged: Luristan material is largely unprovenanced	Cast bronze for horses: the steppe link
WS-092	Master of Animals cheekpiece, Luristan	c. 700 BCE	Cleveland 1961.33, CC0	Two named owners, no dates. Flagged	Same
WS-093	Amphoriskos, mould-blown glass	50–100 CE, Roman	Cleveland 1919.12, CC0	Two names, no dates; acquired 1919	Blown glass within a century of its invention
WS-094	Janiform head flask, mould-blown	100–400 CE, eastern Mediterranean	Cleveland 1923.945, CC0	No provenance in the record; acquired 1923	Glass as a mould-made product

ID	Object	Date	Holder and licence	Provenance	Role
WS-095	Cup signed "Ennion made me"	c. 1–50 CE	Met 17.194.225; licence to check at record level	—	A named maker
WS-003	Stone mason's chisel	c. 2051–2000 BCE	Met 27.3.12	Excavated	Copper-alloy tools on the Nile (shared)
PL-030	Magan copper sites, Oman (place)	3rd millennium BCE	Published surveys	—	The Gulf's first export

The unprovenanced pattern is the chapter's own lesson: most West Asian antiquities in Western museums arrived through a market fed by looting and by 20th-century wars, and Luristan is a textbook case. The chapter shows the objects with that history and hands the wider argument to F1.24.

Living practice. Gulf boat-building (lenj) and, for Iran, a documented craft such as Kashan carpet weaving or Yazd textile weaving (UNESCO listings to verify). For Mesopotamia, the marsh reed architecture of the Iraqi marshes (a UNESCO World Heritage cultural landscape since 2016, to verify) as a practice that spans five thousand years.

Protocol. No funerary objects or human remains; cult figures shown as cast objects with their use named. The chapter states which countries' export laws the objects' histories are silent about (Iraq, Iran, Syria) and does not speculate about findspots.

The walk

1. **Guess.** What was the first thing anyone wrote down?
2. **Rations.** Barley and emmer on clay, about 3000 BCE: writing begins as accounting.
3. **Copper across the Gulf.** Magan's mines and Sumer's bills: the first documented trade route in F1.
4. **Before the Near East.** Belovode: copper smelted in the Balkans two millennia earlier than the textbook story.
5. **Cast for horses.** Luristan: bronze as harness, and a record that is mostly missing because the ground was dug for sale.
6. **A breath.** Glass-blowing in the Levant, 1st century BCE: a technique that made glass cheap, and a maker who signed his work.

7. Parallel scripts. Egypt's labels and Mesopotamia's tablets: the argument over who wrote first (to F1.5).

8. The reed and the boat. The marshes and the lenj: making that outlasts every empire in the walk.

Apply

Lane	Task
Both	Write the account line for one object in your world: what it is, how many, for whom, where from
Maker	One technique of yours that turned an expensive thing cheap, and what changed when it did
Briefer	Draft the export-and-findspot question a brief must ask of any antiquity-inspired reference

How we will know it works. At least 80% of learners can say what the earliest texts record. At least 70% state that most Luristan bronzes have no findspot and why that matters. Briefers' questions name a country and a law.

Links out. F1.1 (the anchor tablet's sibling), F1.5 (writing, copper), F1.10 (what came before Islam's routes), F1.11 (Rome in the east, glass), F1.13 (the Gulf), F1.24 (Luristan as a looting case), F1.28 (accounts as primary sources), the steppe chapter (horse gear), F1.31 (the Gulf hook).

What changed. A new chapter at Practice depth, about 50 minutes, opening on an object (the ration tablet), Historian voice. Four records are flagged for provenance, two of them F1.1's anchor and its sibling. Two living practices are proposed, both to be verified against UNESCO.

New chapter: the steppe and Central Asia, read through the horse

The horse is the device: the steppe's makers built for movement, and what they made (felt, harness, chariots, portable houses, silk for the road) shaped every settled world around them. The deep dive's main finding is that the steppe's best-known "first" has just been overturned. Botai, taught for a decade as the place horses were first domesticated 5,500 years ago, is now read by geneticists and archaeologists as a hunting site, with modern domestic horses spreading from the Volga-Don steppe about 2000 BCE. The chapter teaches the revision itself.

Reach. From the Pontic steppe to Mongolia, with Sogdiana and the Altai, from the 4th millennium BCE to the present. **Groupings used:** archaeological cultures (Yamnaya, Sintashta, Pazyryk); a diaspora (the Sogdian merchants); a network (the Silk Roads corridor); a technique family (felt, pile knotting, compound-weave silk); polities (the Mongol empire); the environment lens (grassland, high altitude, permafrost).

Claims

Claim	Confidence	Source
Modern domestic horses descend from the western Eurasian steppe, especially the lower Volga–Don region, and spread rapidly across Eurasia from about 2000 BCE together with Sintashta spoke-wheeled chariots; Botai's horses (c. 3500 BCE) were not their ancestors	Documented	Librado et al. 2021, <i>Nature</i>
Botai's horse remains are Przewalski's horses, and the tooth damage read as bit wear is better explained by natural wear; Botai may have been a site of mass hunting rather than domestication	Contested: a strong recent reading against a decade of consensus	Taylor and Barrón-Ortiz 2021, <i>Scientific Reports</i> ; Outram et al. 2009, <i>Science</i> 323, for the earlier view
The Pazyryk carpet, 5th–4th century BCE, from a frozen barrow in the Altai excavated in 1949, is the oldest surviving pile carpet; it has about 1,250,000 symmetrical knots, and where it was made (Iran, Central Asia or the Altai) is unknown	Documented (object); contested (origin)	State Hermitage Museum

Claim	Confidence	Source
The Chang'an–Tianshan corridor linked China to the Zhetysu region from the 2nd century BCE, with posthouses, beacon towers and caravanserais	Documented	UNESCO (shared with F1.13)
Sogdian silks of the 700s carried Persian pearl-medallion motifs and reached China; museum attributions place a single coat as "probably Sogdia" or "China, Tang dynasty"	Documented (object); attribution uncertain	Cleveland 1996.2.1, read live
The Mongol ger is made by households: men carve the wood, women and men felt, sew and paint; it can be dismantled and carried, and is taught by mentoring	Documented	UNESCO, 2013
Luristan's cast horse gear (800–600 BCE) shows steppe horse culture inside Iran	Documented (object)	Cleveland 1980.102 (shared with West Asia)

Objects

ID	Object	Date	Holder and licence	Provenance	Role
WS-100	Pile carpet, Pazyryk barrow 5	5th–4th century BCE	State Hermitage 1687–93; image rights to request	Excavated 1949	The oldest pile carpet; origin argued
WS-101	Child's coat with ducks in pearl medallions	700s	Cleveland 1996.2.1, CC0	A London dealer, 1996; earlier history absent. Flagged	Silk on the road; contested attribution shown
WS-102	Roundels with hunters, samite	800s, eastern Iran or Central Asia	Cleveland 1959.124, CC0	A Paris dealer, 1959	Compound weave; riders as motif

ID	Object	Date	Holder and licence	Provenance	Role
WS-091	Horse bit, Luristan	800–600 BCE	Cleveland 1980.102, CC0	Flagged	Harness as art (shared)
WS-103	Sintashta chariot burial	c. 2000–1800 BCE	Excavation photographs; rights to request	Excavated	The spoked wheel
WS-104	Ordos-style animal plaque	1st millennium BCE	Not in Cleveland's CC0 set; the Met at record level	—	Cast bronze on the move
PL-040	Botai (place)	c. 3500 BCE	Published sites	—	The revised first
PL-041	The lower Volga–Don steppe (place)	c. 2200–2000 BCE	Librado et al.	—	Where domestic horses spread from

The Hermitage and the Russian and Mongolian state collections hold most of the steppe's great objects, and none of them publish under open licences. The chapter will need image permissions or a reliance on excavation photographs, which the harvest plan should budget for.

Living practice. The Mongol ger: felt-making, wood-carving and the annual dismantling and raising, with herders' own accounts. Kazakh and Kyrgyz felt carpets (*shyrdak*, UNESCO 2012, to verify) as the second.

Protocol. Pazyryk and Sintashta are burials; the objects are shown, the bodies are not. Attributions that put one coat in two countries are shown as the museum's uncertainty, not resolved by us.

The walk

1. **Guess.** Where and when were horses first tamed?
2. **The revision.** Botai in the textbooks; Botai in the genetics; the Volga–Don about 2000 BCE. A first that moved.
3. **The wheel and the horse.** Sintashta: spoked chariots and paired horse burials; horses and Indo-Iranian languages moving together.
4. **Knots in the ice.** The Pazyryk carpet: 1,250,000 knots, and a maker no one can place.
5. **Harness as art.** Luristan's cast bits: the steppe's horse in Iran's mountains.
6. **Silk for the road.** The Sogdian coat: pearl medallions from Persia, worn towards China; the corridor's posthouses.

7. **A house that folds.** The ger: felt, wood and rope, made by a household, taught by watching.

8. **What the steppe gave the settled worlds.** Riding, felt, the stirrup and the composite bow, listed with their best-dated evidence, and the caveat that each is a "first known".

Apply

Lane	Task
Both	Take one object in your world and design it to be carried: what changes in material, joints and weight
Maker	Name one "first" in your field that has been revised, and what the revision changed
Briefer	Write the review clause a brief should carry for any historical claim: who checks it, against what, and when

How we will know it works. At least 80% of learners can say why Botai is no longer the answer. At least 70% list two steppe contributions to settled making with a dated source. Briefers' review clauses name a checker and a date.

Links out. F1.5 (a revised recurrence), F1.13 (the corridor), F1.14 (motifs copied across Eurasia), F1.18 (felt-making as women's and men's work), F1.24 (frozen tombs and who owns them), F1.28 (genetics as a source), F1.30 (the steppe's own calendars), West Asia (Luristan).

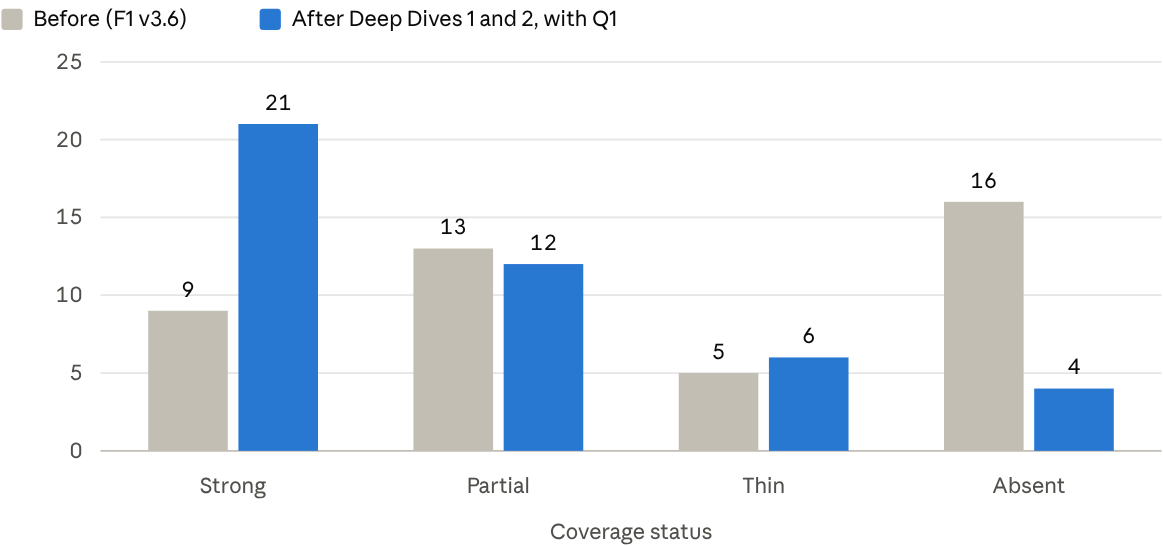
What changed. A new chapter at Practice depth, about 50 minutes, opening on a prediction (the guess), Historian voice, with the Botai revision as its spine. Its objects are mostly in collections without open licences, which the harvest plan must budget for. One Cleveland record is flagged.

Coverage after this round

With the two new chapters and the widened worlds, the map is close to whole. Of 43 sub-regions, 21 are now covered strongly, 12 partly, 6 thinly and 4 not at all. The four still absent are the Northwest Coast, the Eastern Woodlands, Madagascar and the Indian Ocean islands, and the Himalaya, all logged for the world hubs. The thin six (Byzantium and the Balkans, Scandinavia and the Sámi, the Caribbean, Central America, Korea, the Maghrib) each have one object or one claim and need a second.

Absent sub-regions fall from 16 to 4 after the second round of deep dives

Sub-regions of 43 by coverage status (count)



the 43-sub-region grid from Deep Dives 1 §2, recounted by hand after this round · 28 September 2026

Time. The period grid is more even than before. Deep time (before 8000 BCE) now sits in F1.5, F1.12 and the steppe chapter; the first millennium BCE and the first millennium CE, which were thin, are carried by West Asia, F1.11’s antiquity and F1.13. The 20th century is still carried mainly by living practices, which is by design: the thematic units F1.19–F1.22 own that century.

What the object checks found across the round

Finding	Count	Effect
Records read live and kept as exemplars	46 across the nine units, 31 with provenance in the record	The taught sets are real
Flagged for provenance	13, including F1.1’s anchor tablet and its sibling	A provenance pass before G3
Excluded	1 (the Belitung cargo)	Taught as a case
Units with no licensed Ring 1 object yet	F1.12; the steppe chapter	Harvest from Te Papa, Australian state collections, the Smithsonian; image permissions from the Hermitage

Finding	Count	Effect
Pipeline fault	The Met's public-domain search filter returns non-public-domain objects	Every hit checked at record level, as the template now requires

Groupings named across the round. 9 networks, 14 polities, 8 archaeological cultures, 12 maker communities, 6 belief traditions, 3 horizons, 2 diasporas, 1 institution (the waqf), 1 movement (Mingei) and 8 civilisation names that will open onto the groupings beneath them. This is the seed list for the grouping pages in Atlas Logic §6.

Decisions, register entries and next steps

#	Decision	Recommendation
Q13	The two new chapters' devices	West Asia before Islam read through the account; the steppe read through the horse
Q14	Where the new chapters sit	West Asia as F1.9a after South Asia; the steppe as F1.12a after the Pacific, so the two worlds that link Asia to Europe close the world chapters before the networks unit; hours rise from 4 / 22 / 22 to about 4 / 24 / 22
Q15	The Botai revision as F1's model of a changed "first"	Teach it in the steppe chapter and reference it from F1.5 and F1.28
Q16	Object sources beyond Cleveland and the Met	Add the Rijksmuseum, Te Papa, the Smithsonian's open collections and Australian state collections to the harvest; budget for image permissions from the Hermitage and national museums for the steppe and Pacific
Q17	State-media and institutional sources	Allowed when paired with an independent or maker's own account, and labelled (the Jingdezhen rule)

#	Decision	Recommendation
Q18	Cleveland's "-style maker" and "unknown female" attributions	Kept as the museum's wording beside our reading, on every card (extends Q2)

Register entries from this round

Register	New entries
World set	WS-020 to WS-104 as listed in each unit: 46 records kept, 13 flagged, 1 excluded (WS-X05)
Claims	About 70 across nine units, each with source depth and confidence
Practices	PR-010 wayfinding; PR-011 lenj; PR-012 Arabic calligraphy; PR-013 Diné weaving; PR-014 Jingdezhen; PR-015 Jamdani; PR-016 compagnonnage; PR-017 the Mongol ger; PR-018 Budj Bim aquaculture
Place nodes	Madjedbebe, Budj Bim, Kilwa, the Chang'an-Tianshan corridor, Fustat, Sripuranthan, Dhaka, Arezzo, La Graufesenque, Paris, Uruk, the Magan copper sites, Botai, the lower Volga-Don, Pazyryk, Sintashta
Documents	DOC-001 Ben Yiju's accounts; DOC-002 the <i>Yingzao Fashi</i> ; DOC-003 a Dutch cloth order; DOC-004 the <i>Livre des métiers</i> ; the 1721 Act
Datasets	SlaveVoyages; the Open Khipu Repository; Digital Benin; PeriodO; Cliopatria
Link register	Each unit's links out, above
Pipeline findings	The Met search-filter fault; the rate limit; no-open-licence collections for the steppe and the Pacific

Next steps, in order

1. Your answers to Q7–Q18 (Atlas Logic and this document), and to Q1–Q6 and P1–P11 if not yet given.
2. The provenance pass on the 13 flagged records, reading each museum page.

3. The harvest: about 60 Ring 1 records in Cleveland and the Met at record level, then the Rijksmuseum, Te Papa and the Smithsonian for the textile and Pacific gaps.
4. The Historian pre-read of all nine world chapters as a regional specialist, then the reciprocal reviewers; the paid reads for F1.13, F1.12 and F1.6 when budget exists.
5. The grouping pages: the seed list above, written to the Atlas Logic page design, starting with Mingei, Nok and the Igun Eronmwon guild.
6. The standard and light specs for the remaining 26 units, in batches, then G3.